

PABLO UBILLA PAVEZ

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EDUCATION

PhD	Machine Learning — INRIA & Université de Montpellier
2025–Now	Thesis on <i>Deep Learning Methods for Species Distribution Models</i> . Supervised by Diego Marcos & Christophe Botella.
MSc	Ecology and Data Science — University College London
2023–2024	Degree awarded with Highest Honours (Top 5%). Thesis on <i>Consensus Clustering for Functional Groups</i> . Supervised by Daniel Maynard.
Master	Operations Management — Universidad de Chile
2022–2023	Degree awarded with Highest Honours (Ranked 1/15). Thesis on <i>Outlier Detection for Elections with Demographic Groups</i> . Supervised by Charles Thraves.
Prof. Degree & BSc	Industrial Engineering — Universidad de Chile
2017–2023	Degree awarded with Highest Honours (Ranked 1/174). Exchange period at Universiteit Leiden (Spring 2022).

RESEARCH INTERESTS

Machine Learning, Statistics, and Ecology.

PUBLICATIONS

Journals

2026	Ubilla Pavez, P., Hermosilla, D. & Thraves, C. <i>An accurate, fast, and scalable ecological inference algorithm for the $R \times C$ case</i> . Statistics and Computing, 36, 195. doi:10.1007/s11222-026-10946-1
2026	Ubilla Pavez, P., Paz, A., & Maynard, D. S. <i>Functional group classification using consensus clustering</i> . PLOS Computational Biology, 22(5), 1–25. doi:10.1371/journal.pcbi.1014278

Workshops

2026	Menoud, A., Servajean, M., Ubilla Pavez, P., Larcher, T. & Marcos, D. <i>On the value of images to learn ecology-aware geographic embeddings</i> . ICPR 2026 Workshop (CVBMC).
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SOFTWARE

faitei: Ecological inference methods (R package in CRAN, 2025). Work with Thraves, C., and Hermosilla, D.

WORK EXPERIENCE

Oct 2024 – Sep 2025	Airnguru, Data Scientist
Remote, Chile	Developed automation of large-scale pricing strategies (up to billions of fares).
Jan 2021 – Jan 2022	Complex Engineering Systems Institute, Research Assistant
Santiago, Chile	Adaptive question selection using dynamic programming, greedy algorithms, and reinforcement learning to maximize the information obtained at the end of a survey.
Sep 2022 – Jan 2023	Municipalidad de Vitacura, Data Scientist
Santiago, Chile	Estimated daily street-level recycling collection and optimized truck routes using GPS historical data, leading to a redesigned partition of municipal collection sectors.
Aug – Oct 2020	Fundación Tierra Viva, Intern
Santiago, Chile	Contributed in data collection for <i>Resumen del Estado del Medio Ambiente 2020</i> .

TEACHING EXPERIENCE

*Seasons correspond to the hemisphere of the institution. †TA: Teaching Assistant; EA: Exam Assistant; I: Instructor.

FRB-CESAB

June 2026	Introduction to Deep SDMs (I), AI for Ecologists: a toolkit
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University College London

- Fall 2023 **Financial Engineering** (TA), MSc Computational Finance
 Spring 2024 **Applied Computational Finance** (TA), MSc Computational Finance

Universidad de Chile

- Fall '20, '23 **Modelling and Optimization** (TA), Industrial Engineering
 Spring '21, '22 **Decisions Under Uncertainty** (TA), Industrial Engineering
 Spring 2021 **Game Theory and Strategy** (TA), Industrial Engineering
 Fall 2021 **Probabilities** (TA), Industrial Engineering
 Spring '20, Fall '21 **Operations Research** (TA), Industrial Engineering
 2019–2022 EA for **6 different courses** in Industrial & Mathematics Engineering

Universidad Finis Terrae

- Fall 2023 **Stochastic Models** (I), Industrial and Informatics Engineering

Universidad Adolfo Ibáñez

- Spr. '21, Summ. '22 **Operations Research** (TA), Industrial Engineering

AWARDS & SCHOLARSHIPS

- 2024 **Best Presentation Prize**, MSc Ecology and Data Science UCL
 2024 **Dean's List**, MSc Ecology and Data Science UCL
 2024 **Academic Distinction Award**, Engineering UCH — awarded to one graduate per Master's
 2023 **Masters Abroad Scholarship**, ANID, national competition (Chile) — full tuition and living stipend
 2022 **Academic Excellence Scholarship**, MSc Operations Management
 2022 **International Mobility Scholarship**, Santander Bank
 2020 **Best Teaching Assistant**, Industrial Engineering UCH — voted by students
 2018–2021 **Outstanding Student**, Faculty of Physics and Mathematics UCH

TALKS & POSTERS**Deep Learning Integration for Species Distribution Models**

- 2026 **World Biodiversity Forum** (*Davos, Switzerland*) (Talk)
 2026 **DeepSDM Day** (*Sion, Switzerland*) (Poster)
 2026 **The European Conference for Biodiversity Monitoring** (*Montpellier, France*) (Talk)
 2026 **Rencontre CiSStats-RESSTE** (*Avignon, France*) (Talk)

Consensus Clustering for Functional Groups

- 2026 **INRIA PhD Seminar** (*Montpellier, France*) (Talk)
 2024 **British Ecological Society: Macroecology** (*Cardiff, UK*) (Talk)

A Fast Ecological Inference Algorithm for the RxC case

- 2025 **Workshop in Management Science** (*Pucón, Chile*) (Poster)
 2024 **Operational Research Society 66** (*Bangor, UK*) (Talk)

Optimized sensor deployment in biodiversity monitoring

- 2024 **Surrey Wildlife Trust Showcase** (*Surrey, UK*) (Talk)

DEVELOPMENT

- Summer 2026 **Probabilistic AI School** — Probabilistic AI Collective
 Vilnius, Lithuania Probabilistic modelling; variational inference; deep generative models.
 Summer 2026 **Probability meets Biology III** — University of Bath
 Bath, UK Collaborative 4 day workshop between biologists and mathematicians.
 Summer 2023 **XVIII Summer School in Discrete Mathematics** — ISCV
 Valparaíso, Chile High-chromatic graphs; provable algorithms for machine learning; linear programming.

SKILLS

Programming	Python, R, SQL, MATLAB	Machine Learning	TensorFlow, PyTorch, Scikit-Learn
Optimization	Gurobi, OR-Tools, CVXPY	Other	Git, L ^A T _E X, HPC